GRADEOPS

Grading handwritten exams is time-consuming, inconsistent, and prone to fatigue-induced bias. To solve this, develop a Human-in-the-Loop (HITL) grading pipeline that uses Vision-Language Models (VLMs) and Agentic LLMs to evaluate scanned exams against strict rubrics.
The system will allow professors to upload bulk exam scans and JSON rubrics. It will automatically extract, transcribe, and grade complex handwritten answers, awarding partial credit with justifications. These AI-proposed grades are then pushed to a high-speed dashboard, allowing Teaching Assistants (TAs) to rapidly review, approve, or override the decisions, ensuring fairness and massive time savings.

GOALS:

- A web portal to upload bulk exam scans (PDFs) and define granular grading rubrics.
- Role-Based Access Control (Instructors vs. TAs).
- Specialized OCR/Vision models to extract messy handwritten answers and store them in cloud storage.
- An Agentic LLM pipeline to:
- Award partial credit based on strict rubric conditions.
- Generate structured textual justifications for the score.
- Flag highly similar logic structures across papers for plagiarism.
- A high-throughput review dashboard showing the cropped student answer side-by-side with the AI grade, featuring keyboard shortcuts for rapid TA approvals or overrides.

TECHSTACK/FRAMEWORKS :

- Machine Learning : Python, PyTorch, Hugging Face (Nougat/Qwen-VL for OCR), Langchain and Langgraph.
- Web : HTML/CSS + React.js, FastAPI or Node.js, PostgreSQL or MongoDB.

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